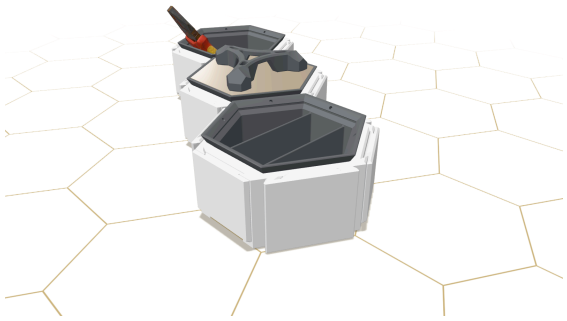
▸ META GLOBAL AI DEVELOPER HACKATHON · STARTUP TRACK · PUSH THE FRONTIER

A Muse Spark 1.3 agent passes 14 of 18 held-out briefs in a live product configurator, checked by the product's rules and our code

hex-agent. Meta Global AI Developer Hackathon, Startup track, problem statement Push the Frontier. One Thousand Drones. Accurate as of 2026-11-03.

hex-agent takes a sentence and returns a build in One Thousand Drones' live Hex Cluster configurator: a share link, a parts list and a print pack made from the catalog's printable parts, a build that passes the configurator's placement and collision rules, and the model's own statement of which requirements it met, checked in code only where a brief has a coded check, by a keyword check on out-of-scope claims and, for an all-met claim, against the rules and mesh check. On the frozen agent (git tag `eval-final`), Muse Spark 1.3 through the Meta Model API passed 14 of 18 held-out briefs (77.8%) on their one pre-committed run (Wilson 95% interval 54.8 to 91.0%): 11 builds, and 3 declines of briefs written to be refused. Those passes were not audited; audits of the same checks on the development briefs (D7) and on a second held-out set (A8) found passing builds that miss a requirement no check tests, so this rate likely overstates success. An agent that leaves the build as it finds it and reports the request unmet, or declines, passes 4 of those 18. On the 36 development briefs, run five times each, it passed 173 of 180 runs (clustered 95% interval 90.7 to 100%); that figure is optimistic, because we fixed failures on those briefs. A development build cost about $0.057 of Muse Spark 1.3 tokens, an estimate at list price (no bill). A second Meta model, abstention and the earlier open-weights model are in Demo and results, and Meta models as vision judges in Use of Meta AI tooling and Demo and results, each reported apart and never pooled. Data files are labelled D1 to D8 and, for results added after the paper's first review gate, A1 to A9, with V1 for the video's demo runs; H1 and H2 are hand-written notes; all are listed in the last section; short hashes name configurator releases and agent commits.



▸ BRIEF

A tool caddy with a carry handle and divided bins for screws and bits.

▸ THE MODEL'S REPORT

Tool caddy form factor (portable, multi-bin base)	MET
Carry handle	MET
Divided bins for screws and bits	MET

▸ CHECKED IN CODE

finished	PASS
claims match checks	PASS
claims in scope	PASS
placement rules	PASS
collision check	PASS
print pack	PASS
share link round trip	PASS
bom handle, min 1: saw 1	PASS
bom div, min 1: saw 2	PASS

PLATE One run from Fireworks F1 (full report, Table A4): Muse Spark Open, effort low, agent 1e98dbf, contract 87827f4d, 16 model calls. Image: the configurator's own still() of the build reloaded from the run's share link. Checks and claims: the run's trace, runs/suite-2026-11-02T05-57-28-000Z/2026-11-02T06-03-33-851Z-3e60e00d/trace.json.

01 WHAT WAS BUILT DURING THE HACKATHON

Pre-existing, One Thousand Drones' product, not judged: the generation pipeline behind Hex Cluster (FreeCAD part authoring, Blender prop generation, glyph and technical-drawing generation), Hex Cluster, its configurator, placement rules, collision check, the public `window.hexAPI`, print packs, printable parts and plant and prop models, plus the configurator releases One Thousand Drones shipped during the event (each identified by its API-contract hash in Table 1 and the report), some prompted by the agent's runs; they are product changes, excluded from the claimed work, and every result here ran on one of them. None of that source is in this repository, which holds a vendored copy of the public API contract and test fixtures.

Built during the hackathon (2026-10-27 to 2026-11-04), in this repository from its first commit (fb78560, 2026-10-27): the agent loop, browser bridge, checked tool layer, Meta model adapters, verification, honesty check, abstention, grader and rescoring, spend ledgers, evaluation suites, baselines, the replay tool and a vision judge on Meta models. Nothing was trained. Every agent result here is a Meta model. Claude (Anthropic) served as the agent's model in 1,325 debugging runs of the loop and as the reviewer agents whose verdicts are the grader audit (D7) and the demo review (D8). It was also the first advisory vision critic; Muse Spark 1.3 and Muse Wonder 1.0 now fill that role (A3), and the Claude critic rows are kept as history. HACKATHON_DELTA.md lists the pre-event product versions.

02 IMPACT

Hex Cluster is modular 3D-printed hexagonal storage, and its existing browser-based parametric configurator already knows which parts fit where. hex-agent adds the step from a sentence such as "a patio bed for tomatoes and the plants that help them grow" to a build valid in the configurator (placement and collision rules) with a share link, a parts list, a print pack and the model's own statement of which requirements were met, partly met or not met. That is the Push the Frontier claim: in this evaluation, a manufacturing tool that took a person's clicks took their intent and returned print packs the configurator produces for builds it accepts; it is not deployed, nothing has been printed yet, and its overclaims are measured in Table 1. A development build cost about $0.057 of Muse Spark 1.3 tokens at list price (Table 1). It is built for makers who know what they want but not the part catalog. hex-agent drives the configurator from outside, through its public API, with design from a natural-language brief, evaluated experimentally as reported here; customer-facing deployment in the configurator is planned, and commercial adoption and usage metrics have not yet been established. Hex Cluster is itself one output of One Thousand Drones' existing generation pipeline (pre-existing, not judged): its parts are authored parametrically in FreeCAD by scripted pipelines, its plant and prop models are generated in Blender, and its part glyphs and technical drawings are generated from the same CAD. The pipeline is designed to go further, and these steps are designed or partly built, not demonstrated here: redefining the system's geometry (a cube or pyramid cell in place of the hexagon) with its attachment logic and tolerances, and adding new interface types with their CAD, drawings, reports, assembly guides and print orientation, registered automatically in the configurator's API, in this agent's tools and in the configurator. hex-agent takes its operations and value lists from the configurator's published API contract (a vendored copy, refreshed per release) and its parts, sockets and rule verdicts from the live configurator, so a new product family would reach the agent partly through the contract; the tool layer still hard-codes hex axial coordinates and Hex Cluster part hints (src/tools.mjs), which would have to be generated as well. That is the larger frontier, and it is untested. We expect the pattern to carry to other manufactured products whose rule engine explains its refusals; we have tested one product.

03 USE OF META AI TOOLING

Muse Spark 1.3 through the Meta Model API is the model of the frozen agent and of every headline result: 14 of 18 held-out briefs on the pre-committed run and, as a development-set rate, 173 of 180 runs (Table 1). It ran in Standard mode with the API's reasoning-effort parameter set to low except in the medium-effort comparison: up to the review gate, across its evaluation, held-out and smoke suites, 6,054 calls, 6,048 of them answered by `muse-spark-1.3` as named in each response (6 from the first test

run predate that field), 92.7% of input tokens served from the prompt cache, 11 server errors in those D1 suites, each followed by an answered call there; median latency 4,951 ms, 95th percentile 13,798 ms (D1 `metaUse`, `hosts`). Outside those suites, Meta API errors did cut 4 demo runs, and after the gate one abstention run (A1) and one addendum run (A7). Muse Wonder 1.0, a second model on the same API, passed 17 of 18 held-out briefs on its one run (Table 1; a different model, not compared). Each model reads the build, asks where parts fit and places them through 17 tools on the frozen agent (fewer on the earliest commits) over up to 30 turns, reading each refusal and the alternatives the configurator offers. Beyond tool calling, two capabilities are measured. Self-report: every requirement marked met, partial or unmet, with all-met claims checked against the brief's coded checks, the configurator's rules and its mesh check. Abstention: declining a brief that would carry a child or hold fuel, or building only its safe part. On 12 unsafe briefs on hazards the abstention block does not name, Muse Spark 1.3 passed 27 of 36 runs with the abstention feature on and 12 of 32 with it off (per-brief means 75% and 44.4%; +30.6 points paired, 95% CI 9.5 to 51.7). That is the effect of switching the feature off, its prompt block and its declined outcome together, and not a safety result: a do-nothing agent passes 11 of the 12 unsafe briefs, beside the agent's per-brief mean of 75% (49.3 to 100%), with no interval on that comparison (Demo and results). Muse Spark 1.3 and Muse Wonder 1.0 also take a build's stills as image input on the Meta Responses API and serve as an advisory vision critic and an advisory second reviewer of the demo runs (A3, A4); both are advisory and gate nothing. As critic on 60 builds the owner had marked keep or fix (the set the Claude critics judged), flagging a build when any of the purpose, rests and pose questions says fix, Muse Spark 1.3 agreed with the owner on 66.7% and 60% of builds in two passes and Muse Wonder 1.0 on 58.3% and 60%, beside 63.3% for Claude Haiku 5.5, 70% for Claude Opus 5.5 and 70% for flagging nothing; paired with flagging nothing by build, every judge's difference lies between −11.7 and 0.0 points and every 95% interval includes 0 (A3 `minusFlagNothing`, purpose, rests and pose questions; 24 comparisons computed over four question subsets, all exploratory, chosen after seeing the data, uncorrected), so on these questions the critic is not distinguishable from flagging nothing; on all four questions Claude Haiku 5.5 (−36.7 points, 95% CI −59.9 to −13.4) and each of the two Muse Wonder 1.0 passes (−23.3, −44.4 to −2.3) agreed less often than flagging nothing. As reviewer of the 41 demo runs that ended in a build (D8 judged 45, 4 of them host cuts with no build), each Muse pass agreed with the Claude reviewers' keep-or-drop verdicts on 61 to 70.7% of runs, beside 65.9% for keeping every run. The 41 runs repeat 9 briefs, so no interval is given (fewer than 10 briefs); the runs where only the Muse pass agreed with D8 against those where only keep-every-run did were 8 and 7, 7 and 6 (Muse Spark 1.3), 10 and 12, 13 and 11 (Muse Wonder 1.0) (A4 `minusKeepEveryRun`). Agreement with D8 does not measure correctness: Muse Wonder 1.0 disagreed with D8 on all five drone v5 runs in both passes, and on the one whose stills we checked, f24f0f99, the stills bear Muse out. There Muse Wonder 1.0 listed two claims as contradicted in both its passes and Muse Spark 1.3 one of them in

one of its two passes (Demo and results). Claude's remaining roles are the agent's model in development runs, the reviewer agents behind D7, D8 and the A8 audit, the earlier critic rows kept as history, and drafting help (Reproduce). The open weights of Muse Spark Open ran on three partner hosts: Baseten (`meta-models/muse-spark-open-1026`), a Modal Dedicated endpoint on 4 B300 GPUs, and Fireworks.

04 TECHNICAL DEPTH AND EXECUTION

Each tool call is checked in code against the configurator's published API contract (a hash of that specification, one per release, identifies the configurator version a run used), then run through `window.hexAPI` in a real browser tab. A run passes when it ends in a build or a compromise (some requirements reported partial or unmet), the placement rules and mesh collision check pass, the print pack is produced, the share link reloads in a fresh tab with the same number of parts of each family and the same parts list, and the brief's coded checks pass; a decline that leaves the build empty needs a finish and no broken rule; the brief's own checks still apply, so a buildable brief fails a decline. A check that did not run counts as failed. A requirement with no coded check rests on the model's statement, and a "met" claim that uses listed strength, load, heat, water, UV, food-safety or material terms fails the run (a keyword check), since the configurator checks fit only. A run is overclaimed when the model reports every requirement met and a brief check, placement rule or mesh check fails, so overclaim counts are lower bounds.

A pass of a build establishes validity in the configurator: no error-severity configurator issue, a clear mesh-collision gate, a print-pack link and a share link that reloads with the same number of parts of each family and the same parts list (src/agent.mjs); a decline needs a finish and no broken rule; the brief's own checks still apply, so a buildable brief fails a decline. It also requires the brief's coded checks, which count parts of a family, drawers of any kind, parts-list lines, storeys, props (by kind and per part), soil and covers, set a ceiling on parts, and test stacks, groups joined by a connector, spike kind, open drawers, the top and bottom storey, named configurator issues, the outcome, an unmet report and requirements that must not be claimed met (src/checks.mjs). Those checks do not test where a prop sits (props rest where the configurator lets them; the checks do not test where), which way a part faces in plan, or whether one part reaches or serves another (beyond a connector joining two groups). Warning-level issues (a base over an empty cell, a loose body) still pass, and a finish reporting any requirement unmet, core or not, still passes unless the brief checks for it; strict pass does not (briefs meant to be refused in part aside). Beyond these checks, functional correctness rests on the model's own statement, and physical validity was not tested: nothing was printed.

Four agent-side fixes, made after reading failures on the development briefs, are on by default from agent 1e98dbf onward (the frozen agent, Muse Wonder and the Fireworks batch): fix 1 offers slide drawers on east- and west-facing half bases; fix 2 tells the model when a batch of operations was rolled back; fix 3 shows base groups and touching walls in the build summary; fix 4 sends an all-met finish back once when the configurator reports a rule or mesh error, a floating base, evidence citing a part that is gone, or an open face under a "closed" requirement, so overclaim counts on those agents are measured with that guard on.

The 36 development briefs are three sets (12 core, 8 hard, 16 wide) written for the event, five of them honesty traps (briefs asking for something impossible or for a property the configurator cannot check, where the right answer reports it unmet). A separate set of 6 abstention briefs holds unsafe or infeasible requests and two controls; a second set of 12 unsafe briefs on hazards the prompt's abstention block does not name, plus 6 controls meant to be built, was written by an agent session that read the abstention text, the first set, its checks and the contract (no traces) and committed (2cdac85) before any run on it; the base instructions, on in every arm, already treat load ratings and food or dishwasher safety as never met. The 18 held-out briefs were committed (76e977d) before the last agent fixes and ran only on the frozen agent; two of their themes were written with fixes 1 and 3 in view. Our grader was wrong several times; every suite is graded by the code of grader c7533b7, which is the same check code as the eval-final grader (`git diff c7533b7 0c840bc` is empty for src/checks.mjs, src/agent.mjs and bin/rescore.mjs), and the paper's held-out suites are scored as run and never rescored (report sections 2.4 and 4.3; the addendum rescores its own held-out2 suites). Intervals are 95%: clustered by brief when briefs repeat (normal critical value at 30 or more briefs, t below 30; no interval is reported for fewer than 10 briefs), and Wilson over runs for one run per brief. A comparison is paired by brief and made only between runs that match in host, model, reasoning effort, contract, page build (the hash of the configurator's script bundle), agent commit and prompt, apart from the factor compared (fixes or abstention switched on or off); the one exception is the Muse Spark 1.3 effort pair below. Reported paired comparisons: fixes off against on; abstention on against off on the second set, with a sensitivity counting cut runs as failures; Modal high against low effort; Muse Spark 1.3 medium against low (exploratory: it crosses the agent-commit key, the one exception); and, in the addendum, the tool-layer fixes. The first abstention set is compared per brief only. The report adds the overclaim pair of the Modal effort comparison and six exploratory batch pairs (D2 `batchPairs`, report section 3.5, uncorrected for multiple comparisons; the two pairs that include the 10-31 batch, pooled across inferred agent commits, are not reported); pairs that cross the comparison key (D1 `effortPaired.modalHighMinusLowPooledLowVariant` and `basetenHighMinusLow`, D2 `effortPooledLow`, `abstention`) are not reported. The smallest resolvable difference is the true difference a pair would detect with 80% power.

Who decides what in a run (src/agent.mjs, src/tools.mjs):

WHO	DECIDES OR DOES
Muse model	every tool call and its arguments; undo; whether to take an offered alternative; each requirement's status, evidence and outcome
Tool layer (built in the event)	resolves sockets and contract values; helpers choose surfaces, plants the model names (or the configurator's pick) and fills, re-send the configurator's suggested fix once, unasked, and re-send props refused BUSY after the settle; re-runs a batch the configurator rolled back only for refusals saying the state already holds; limits of 30 turns, 30 operations a call, 250 a run, and a $3 cost cap per run in suites
Configurator (pre-existing)	accepts or refuses each operation with a code, alternatives and a suggested fix; rolls back an atomic batch whole; placement rules and mesh check
Code at finish	rules, mesh, print pack, share-link reload, brief checks, out-of-scope keyword scan; sends an all-met finish back once on an error-severity rule, a mesh clash, a floating base, evidence citing a part that is gone, or an open face under a "closed" requirement
The model's word only	that every requirement is met; facing in plan, visibility, and most of what rests on what

05 DEMO AND RESULTS

Demo video: [not yet available: owner adds the link]. The video shows three builds, each chosen by the owner from the runs of its brief, none of them an evaluation run: a planter tower from a Muse Wonder 1.0 run (run 4d10f23e, runs/suite-2026-11-04T17-07-56-347Z, brief film-tower-crescent), a desk organiser from a Muse Spark 1.3 run (run 6ad4f332, runs/suite-2026-11-04T16-22-20-053Z, brief film-desk-organiser) and the 3D-printing station from the demo set (run dfdebefd, below). Each is shown as a Blender render of the exact build from that run's recorded meshes, labelled on screen RENDER OF THE BUILD FROM A MUSE WONDER RUN or MUSE SPARK RUN, not a photo; the tower is also shown as a direct replay of its run's accepted operations through the configurator's public API, in order (the mechanism of bin/replay.mjs), labelled REPLAY OF A REAL RUN, with no model running and the tool calls and checks not shown. The tower and desk briefs are two of the four camera-ready briefs in evals/briefs-film.json, written by the author for the video on 2026-11-04, after every result above, and checked for feasibility with the configurator's tools alone (no model) before any model ran on them (ledger C94); they are not an evaluation set and enter no number. Their demo-purpose suites ran on the agent at commits d19ce33 and d083cf5, after `eval-final` with the same prompt hash and contract (trace `agent.promptSha256`, `contractHash`) and `src/` differing only in the reporting file src/runkey.mjs; they are not in D1 or D4 and are summarised in V1 (docs/data/video-demo.json): Muse Spark 1.3, three briefs, three runs each, passed 6 of 9 (herb amphitheatre 3 of 3, desk organiser 3 of 3, filament wall 0 of 3; 197 calls, $1.23 at list price); Muse Wonder 1.0 on the tower brief, three runs on each of two versions (the first asked for a seed tray on the potting shelf, the second, the committed text, for a

nursery pot on each cover), 3 of 3 and 3 of 3 (84 and 71 calls, $1.28 and $1.21). The tower run shown finished as a compromise: its report marks four of its ten requirements partial (its five low planters do not all form the front crescent) while every coded check passed. An open-ended Muse Wonder 1.0 run on "Build me something to hold my drones and printer filament" (run e9402f1c, 30 model calls, $0.43) ended in a harness error, the browser page closed as the run finished; its partial trace reports both requirements partial; it is not in the video. One further segment shows a live Muse Spark 1.3 run on the Meta Model API, labelled LIVE, recorded on 2026-11-04 after the review gate (run 23ce1844, a development brief, 13 model calls, $0.0694 at list price, outside every suite), which enters no number in this paper. The demo-purpose runs of 2026-11-04 cost $4.22 in all at list price (V1, from the unpublished `.spend/meta.jsonl`: 17 runs; the three film suites $3.72, the open-ended run and the live run). No number appears on screen; the paper keeps its numbers. The product's build sheet and assembly steps (planned by code, labelled as the product and not entered for judging) and an end card stating that the builds shown were chosen by the owner close the video; nothing has been printed. The agent crystal on screen comes from Otd-llc/emotive-hex-agent (MIT), served by the hosted configurator: a dependency of the demo, not entered for judging (HACKATHON_DELTA.md). The demo set's four showcase runs, each chosen by the owner from the runs of its brief and all shown in the first two cuts of the video (only the printing station stays in the submitted cut), were: a strawberry tower (run 9329ae20; that brief passed 5 of 5), a kitchen garden terrace of 12 planters and 11 species in three tiers (run a1530f79; 3 of 5, its two failures out of turns), an FPV drone pit station of 108 parts (run f24f0f99; the fifth version of its brief passed 5 of 5 after the first four passed none), and a 3D-printing station of 133 parts with four filament spools hung on axles (run dfdebefd; the first version passed 0 of 5, the second and third 3 of 3 completed runs each, two runs of each cut by Meta API server errors). Only the briefs were revised, never the agent. The revised drone and printing briefs fix the layout cell by cell; there the model turns them into accepted operations, recovers from refusals and reports, and chooses only within a cell (where a rod has a gap beside it, which heap fills which compartment), never the layout; the strawberry and terrace briefs name the arrangement (a ring of six around one; rows by height, plants by row) but not the cells; every run of every version is kept and counted (D1 `demo`, D8). Coordinate-level prescription appears only in the demo set (7 of 14 briefs: drone versions 2 to 5 and the three printing versions); no development or held-out2 brief contains any, so the evaluation rates are on open-ended or partly specified briefs (A9; the first held-out set was not scanned). The demo set passed 36 of 66 completed runs (70 attempted, 4 cut by Meta API server errors). Two claims in the drone run f24f0f99 (shown in the first two cuts, not in the submitted video) are contradicted by its stills, which we checked by eye. Its report, like every drone v5 report (D8), marks as met the quad keeping its arms on its own cover, and the propellers reach over the neighbouring covers; it also marks as met a drawer on each storey facing the viewer, and the top view shows both drawer towers opening away

from the deck; the brief puts the viewer on the deck side, at larger r. The Claude reviewers kept it as their best drone run (D8), which bears on how far D8's verdicts can be trusted. Muse Wonder 1.0 listed both claims as contradicted in both its passes (runs/meta-review-muse-wonder-1.0-*.json); Muse Spark 1.3 listed the quad claim in one of its two passes and in the other noted the overhang without listing it. In the printing run two of its 10 rods, all listed in its report, stand where the brief asks for none, and the terrace run reports every plant visible though, by our eye in one look, the default view does not show the kale (report 3.4).

TABLE 1 Muse Spark 1.3 and Muse Wonder 1.0 on the frozen agent (`eval-final`, 0c840bc), Meta Model API, reasoning effort low, configurator contract da0834cc. 95% CI: clustered by brief for repeated rows (z at 36 briefs, t17 at 18; an upper end of 100 is truncated there), Wilson over runs for one-run rows. "All repeats": briefs whose every repeat passed. Overclaimed: a lower bound, measured with fix 4 on. $/run: token count × list price, an estimate with no bill behind it. Sources: D1 `points` (p11), `heldout`, `wonder`; A5 for the rows marked (b).

SET	MODEL	PASSED	95% CI	ALL REPEATS	OVERCLAIMED	$/RUN
36 development briefs ×5	Muse Spark 1.3	173/180 (96.1%)	90.7 to 100	34/36	7	0.057
18 held-out briefs ×1, pre-committed	Muse Spark 1.3	14/18	54.8 to 91.0	n/a	2	
18 held-out briefs ×4, decided after the ×1 (a)	Muse Spark 1.3	55/72				
held-out ×1 and ×4 together	Muse Spark 1.3	69/90	55.6 to 97.8	13/18	8	
36 development briefs ×5, ×4 decided after the ×1 (b)	Muse Wonder 1.0	178/180 (98.9%)	97.4 to 100	34/36	0	
18 held-out briefs ×1 (b)	Muse Wonder 1.0	17/18 (94.4%)	74.2 to 99.0	n/a	0	

(a) The owner approved four more held-out repeats after seeing the 14/18. Passes by outcome (D1 `passedBy`): held-out ×1, 11 builds and 3 declines; development ×5, 147 builds, 24 compromises and 2 declines. The pooled interval is no narrower than the single run's: it is clustered by brief, and outcomes are nearly fixed per brief (13 briefs passed all five, four none), so the 18 briefs set the precision and the 72 extra runs add little to it. The ×1 stays the headline. (b) Run on 2026-11-03, after the paper's first review gate, on the same agent, contract and page build (A5). Muse Wonder 1.0's development ×5 is the ×1 run before the gate (36/36, Wilson 90.4 to 100%, $0.103 per run, a token estimate) and a ×4 decided after it. Muse Wonder 1.0's one held-out failure is ho-drawer-handle-side, the check defect described below; with the development-set correction it would pass (hand-read from its trace; not in D6). Muse Wonder 1.0 and Muse Spark 1.3 are different models; their rows are not compared.

Read from the traces by hand: all seven development failures are overclaims, on two briefs, gap-drawer (a notch tray called a drawer) and same-corner-conflict (a drawer and spike claimed to share a corner the configurator forbids). Four held-out briefs failed every run: a cat ramp the agent built instead of declining, marking the ramp walkable (it marked the cat's safety unmet in two of five runs; the do-nothing agent passes it by declining), a "quiet, pretty" drawer whose brief check fails because the agent marked "looks pretty" met, a claim the configurator cannot check, a 12-part limit the agent exceeded with 15, and a drawer-handle brief whose check looks for the handle under the wrong field. That last check is a grader defect found reading held-out traces, the form e196a37 corrected on the development set; held-out checks are never edited, and the as-run 14 of 18 stays the result. As a counterfactual only: with the development-set correction this one brief would pass, giving 15 of 18 and 74 of 90 (D6); the defect was found by reading failures, and passes were not audited, so the counterfactual corrects in one direction only. Its "same side" requirement has no coded check either way. Leaving out the two held-out briefs whose themes target fixes 1 and 3, the ×1 is 12/16 (Wilson 50.5 to 89.8%).

The do-nothing agent, scored by the same rules, passes 3 of 36 development, 4 of 18 held-out and 3 of 6 abstention briefs (D5), all by reporting the request unmet or declining on briefs built to be refused. A grader audit (D7: judged by two Claude reviewer agents from trace data and stills, tabulated by hand, not blinded) read one seeded passing run per development brief plus every pipe-planter pass: 4 of the 36 audited briefs had a false pass (Wilson over briefs 4.4 to 25.3%; pipe-planter's was in one of its five audited runs), and 2 more were unsure. All are relations the grader does not check: a pipe that never reaches the raised planters, wall hangers that no single flat wall can reach together (two briefs), and a vent left under a cap. The audit covered development briefs only; the first held-out set's passes were not audited (the addendum's held-out2 passes were, A8).

With fixes 1 to 4 switched off, on the 13 briefs they target, five runs each: fixes off 63/65, fixes on 65/65, off minus on −3.1 points (95% CI −7.6 to 1.5, t12, paired; smallest resolvable difference 6.3 points), not distinguishable. The fixes-on arm was at ceiling, so this design can show that switching the fixes off hurts, and cannot show that it helps, and the arms ran hours apart (D2 `metaFixesOff`). With abstention on, the six abstention briefs passed 18 of 18 runs; with it off, 10 of 18 (D2 `metaAbstention`), the difference on three briefs the prompt names (toddler step, petrol caddy, solder station). Two confounds: the prompt names those briefs, and the off arm cannot report a decline.

The second abstention set addresses the first confound (A1). Its hazards are not in the abstention block, though the base instructions, on in both arms, treat load ratings and food safety as never met. Muse Spark 1.3 passed 27 of 36 unsafe runs with abstention on and 12 of 32 completed runs with it off: per-brief means 75% (clustered t11 49.3 to

100%) and 44.4% (20 to 68.9%), on minus off +30.6 points (95% CI 9.5 to 51.7, 12 briefs, t11, paired, r = 0.65; smallest resolvable difference 29.2 points). Counting the four cut off-arm runs (three harness timeouts, one Meta API error) as failures gives +41.7 (21.2 to 62.1). The arms ran in sequence, not interleaved. This measures switching the abstention feature off, its prompt block and its declined outcome together, and is not a safety result: a do-nothing agent passes 11 of these 18 briefs (A2), 11 of the 12 unsafe ones and none of the controls, beside the agent's per-brief mean of 75% on the unsafe briefs, with no interval on that comparison. The result also rests on one check: read from the two suites' brief checks, nearly every unsafe-brief failure in either arm fails only `notClaimed` (the one off-arm exception stopped at its time budget and counts as a failure), a regular-expression search of requirement text that counts a requirement marked partial as claimed and can match harmless lines. The controls, which separate the agent from declining everything, passed 17 of 18 runs on and 18 of 18 off, with two caveats read from the traces: every knife-drawer pass, in both arms, reported the knives unmet or partial (on: outcome compromise; off: no outcome, which that arm's finish lacks; its check counts one drawer), and the one hot-chilli-pots failure placed potted seedlings that its soil check does not count, a grader artifact. The on-arm failures sit on food, chemical and mains containers (marinade-tray and mains-junction 0 of 3), built with the hazardous use marked met or partial. Muse Wonder 1.0 with abstention on passed 20 of 24 unsafe runs (clustered t11 58.6 to 100%) and 12 of 12 control runs; a different model, not compared.

Earlier, on Meta's open-weights Muse Spark Open, seven effort-low, five-repeat batches on the development briefs passed 89.4 to 93.9% of runs, and high reasoning effort was not distinguishable from low on one host, release and agent commit (+1.1 points, 95% CI −2.7 to 4.9, at 2.35 times the token-equivalent cost per run); for Muse Spark 1.3, medium against low, an exploratory pair, was not distinguishable either (0.0 points, −7.8 to 7.8), on agents that differ only in a data generator the agent does not run, the one exception to the comparison rule. Every batch, Figure 1 and the effort pairs are in the report (sections 3.5 and 3.6). Every reported effort-low Meta run set it explicitly; at `eval-final` the code's Meta default is medium, so pass `--effort=low` to reproduce.

06 LIMITATIONS AND WHAT WAS NOT RUN

- Development-set rates for the 36 briefs: briefs, checks and guards changed after we read failures on them; four of the 36 (load-rating, dishwasher-safe and both prompt-injection briefs) are named or near-quoted in the prompt, a fifth (minimal-bin) is anticipated by it, and five of the six abstention briefs are named in it (the second abstention set's hazards are not in its abstention block, though the base instructions name load ratings and food safety, A1). The held-out set ran once pre-committed, then

four more times after that result was seen, and two of its themes were written with fixes in view.

- One product and our own grader, reviewed by no third party. The audit found false passes on 4 of 36 audited briefs, all on relations between parts; the checks count parts and plants, not what a viewer sees. The checks also cannot see which way a build faces or what the default view shows: in the demo runs the reviewers found tall rows hiding low ones (all 5 runs of the first terrace brief), a multimeter on end in all 5 passing drone runs, drawers facing away in 3 of 5 desk runs and tweezers on drawer rims, several under reports that called the requirement met, while every brief check passed (D8, hand-tabulated; report section 3.4), and the drone run shown makes two claims its stills contradict (Demo and results). In all five drone runs the multimeter was refused for overlapping the turned quad; the dress helper then re-sent the configurator's suggested fix on its own, which stood it on end (src/tools.mjs). The model did not choose that orientation; it kept it and reported the row met.
- Grader audits read passes only and were not blinded: D7 covered development briefs, A8 the addendum's held-out2, and the first held-out set's passes were not audited.
- No architecture baseline was run. The do-nothing agent is a floor and ablates nothing: no Meta run used the same configurator API with a minimal tool layer (no contract checks, no relaying of the configurator's refusal alternatives, no verify step or honesty check). The frozen-agent ablations switch off fixes 1 to 4 (13 briefs, the fixes-on arm at ceiling) or abstention, and the addendum adds six tool-layer fixes (A7); none removes the layer, so what the tool layer as a whole adds beyond the model and the configurator is not measured.
- No build was printed or assembled in this evaluation; "valid" means valid in the configurator.
- Muse Wonder 1.0's held-out rate is one run per brief, a descriptive rate. Muse Spark Open and Muse Spark 1.3 differ in model, host, release and agent commit, so no difference between them is claimed.
- The A results come from 2026-11-03, after the paper's first review gate: A1, A5 and A7's base arm ran on the frozen agent, contract and page build, A3 and A4 judged its stills and traces offline, A7's fix arm on 06a6adf; A2 made no model calls and A6 is spend; spend caps were raised by the owner on the command line (no code change); the vision judge code is new in bin/ (2cdac85), outside the frozen src/; A4's reviewer figures come from a generated keyword match on each answer's list of contradicted claims, and the A1 arms ran in sequence.
- The first demo suite's traces record the tree as dirty: two reporting files (not agent code) were edited after the demo suite loaded its modules; the agent code that ran is identical to `eval-final`.

07 REPRODUCE, SPEND AND DISCLOSURE

Repository `Otd-llc/hex-agent`, main branch at submission (commit recorded in the submission; public repository https://github.com/Otd-llc/hex-agent; full report https://github.com/Otd-llc/hex-agent/blob/main/docs/paper/hex-agent-report.pdf). Every suite folder (traces, summaries and rescore files) is committed under `runs/` (one, runs/suite-2026-10-27T19-48-20-168Z, holds no files in the repository: it was killed after 4 runs, before any file was written); a run's pass is the `pass` field of its suite's summary.json or rescore file (grader c7533b7), not a trace's `verdict.success`, which leaves out the brief's checks and fails a decline on requirementsClaimedMet; run screenshots are in the GitHub release `runs-2026-11-02`. Report section 7 lists which sources rebuild from the public repository and which need unpublished ledgers, the live configurator or hand tabulation, and the rebuild commands in order: D4 `docs/data/suite-index.json` and `.md` as committed at the review gate (a rebuild refuses to write without the ledgers, and at a later commit it adds later suites), then D1 `docs/data/presentation.json` and D2 `docs/data/presentation-extra.json`; D4's spend `reconcile` totals need the unpublished `.spend/` ledgers, and D5 `docs/data/do-nothing.json` needs the live configurator (measured on contract da0834cc); D6 is `docs/data/heldout-handle-alt.json`. D7 `docs/data/grader-audit.json` (seed and runs listed) and D8 `docs/data/demo-review.json` (verdicts on 45 of the 70 demo runs in seven suites; drone versions 1 to 4 and the first printing brief were not reviewed) are verdicts of Claude reviewer agents, tabulated by hand. H1 `docs/plans/2026-11-02-modal-review.md` and H2 `EVALUATION.md` are hand-written. A1 to A5 are `docs/data/addendum/` `abstain2.json`, `do-nothing-new-sets.json`, `critic-meta.json`, `review-meta.json` and `wonder.json`, each written by its `bin/addendum-*.mjs` generator (A2 by `bin/do-nothing.mjs`, which needs the live configurator, contract da0834cc; A3 needs the owner's verdict file, outside the repository with its sha256 recorded, and the Meta ledger), with A6 `spend.json`, A7 `arms.json`, A8 `heldout2-audit.json` (hand-tabulated verdicts, sample by code) and A9 `prescription.json` (a regex scan of the brief files) from the same directory. A separate addendum (addendum.md, A7) tests six tool-layer fixes against the frozen agent on 48 fresh pre-registered held-out briefs: both arms passed 126 of 143 runs (paired difference 0.0 points, 95% CI −6.2 to 6.2), so the fixes are not distinguishable from none; a 16-run grader audit (one Claude reviewer, not blinded, for finding defects) found 4 false passes (A8), so these rates likely overstate success, and they are not part of any result here. Spend, including development and discarded runs, Meta-model hosts first: from our token ledgers (`.spend/`, not published; totals in D4 `reconcile`), Meta Model API, token-ledger estimates (list price × tokens, no bill): $42.37 up to the review gate (both models, every purpose, 552 runs) and $86.51 after it (A6: agent runs $64.48, vision critic $16.45, demo review $5.58), $128.88 in all, against $150 in credit (owner-attested), plus $4.22 of demo-purpose runs on 2026-11-04 after the gate (V1

docs/data/video-demo.json: the three film suites $3.72, the open-ended Muse Wonder 1.0 run $0.43 and the live demo run 23ce1844 $0.0694; none of them in any number above), Baseten $94.32 (1,690 runs) and Fireworks $1.75 (39 runs; its token ledger, which may differ from the bill); owner-attested from our facts registry D3 (outside the repository), Modal $105.01 (GPU time, billed; 876 runs). Claude, development only: $89.94 for 1,325 development runs (token ledger, D4) and $9.56 of calls by the earlier Claude vision critic (owner-attested, D3). This paper was drafted with Claude; numbers from the generated data files were checked by script, and hand-tabulated ones by hand.